

Identification of Hub Genes in Ulcerative Colitis via Differential Expression a...

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Report Title

Identification of Hub Genes in Ulcerative Colitis via Differential Expression a...

Conversation Name

I have recently obtained microarray expression data from colonic mucosa biopsy...

Report Time

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1. Results

1.1 Identification of Hub Genes in Ulcerative Colitis via Differential Expression and Protein-Protein Interaction Network Analysis

1.2 Introduction and Objectives

Ulcerative colitis (UC) is a chronic, relapsing inflammatory bowel disease characterized by diffuse mucosal inflammation of the colon. Identifying key molecular drivers of active UC is important for understanding pathogenesis and for discovering potential therapeutic targets. In this study, we analyzed a gene-level log2 expression matrix from colonic mucosal biopsies (GSE65114) comprising 16 active UC samples and 12 healthy controls. The objective was to identify hub genes through a systematic pipeline: differential expression analysis with limma, construction of a protein-protein interaction (PPI) network via STRING, module detection, multi-algorithm topological ranking, integration of rankings into a final hub gene list, and biological interpretation of the identified hubs.

1.3 Data Overview and Quality Control

The input expression matrix contained 15,630 genes across 28 samples (29 columns including the gene symbol column). All 437,640 expression cells were numeric with no missing values and no duplicate gene symbols. The sample group table contained 12 healthy controls and 16 active UC samples. All 28 matrix sample columns matched the group table sample IDs, and the column order was identical to the group-table row order, confirming proper sample alignment for downstream analysis.

Metric	Value	Notes
gene_count	15630	Total genes in expression matrix
sample_count	28	16 active UC + 12 healthy controls
uc_sample_count	16	
healthy_control_count	12	

1.4 Differential Expression Analysis

Differential expression analysis was performed using limma (lmFit + eBayes; limma v3.66.0) on all 15,630 genes, contrasting 16 active UC samples versus 12 healthy controls (healthy as reference). Using the criteria $|\log_2FC| > 1$ and $P < 0.05$, 299 differentially expressed genes (DEGs) were identified: 227 up-regulated and 72 down-regulated in UC. The complete per-gene results table (gene symbol, log2FC, P value, adjusted P, DEG status, regulation) was written to outputs/files/deg/complete_deg_list.tsv. A volcano plot and a heatmap of the top 30 DEGs by significance were generated.

Metric	Value	Notes
total_deg	299	$ \log_2FC > 1$ and $P < 0.05$
up_deg	227	Up in active UC
down_deg	72	Down in active UC
genes_tested	15630	

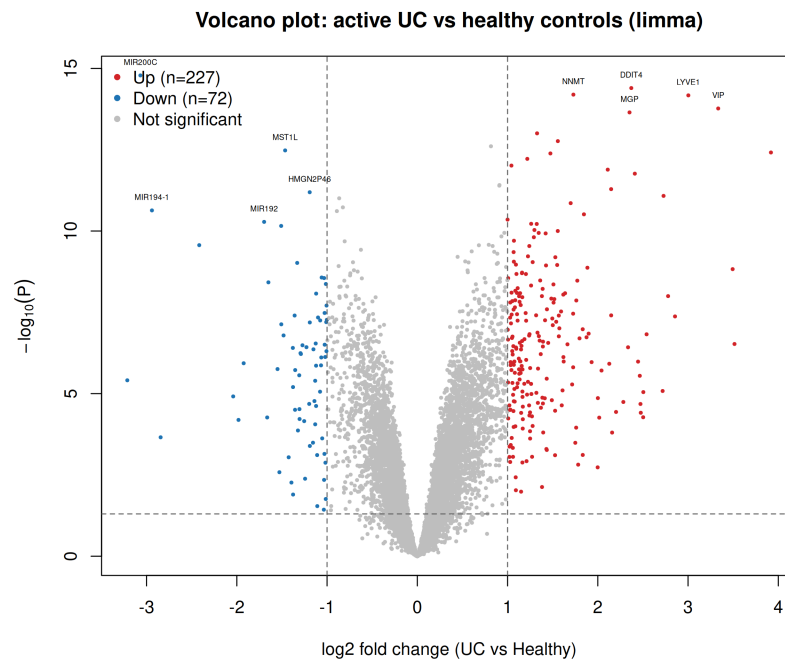


Figure 1 Volcano plot of differential expression (active UC vs healthy controls). Up-regulated DEGs in red, down-regulated in blue, non-significant in grey; thresholds at $|\log_2FC| = 1$ and $-\log_{10}(0.05)$.

The volcano plot shows a clear separation of 227 up-regulated and 72 down-regulated genes passing the $|\log_2FC| > 1$ and $P < 0.05$ thresholds, with many highly significant genes at large effect sizes.



Figure 2 DEG heatmap of the top 30 DEGs by significance, with row z-scores and sample-group color bar (active UC vs healthy controls).

The heatmap shows distinct expression patterns between active UC and healthy control samples for the top 30 DEGs, with clear clustering by group.

1.5 PPI Network Construction

The 299 DEGs were submitted to the STRING database (API v12.0, Homo sapiens taxid 9606) at medium confidence (combined score ≥ 0.400). Of the 299 DEG symbols, 272 mapped to STRING protein identifiers; 27 symbols (mostly microRNAs, lncRNAs, and pseudogene-like transcripts) did not map. The resulting PPI network contained 224 nodes (unique DEG-encoded proteins with at least one interaction at the threshold) and 1,283 undirected edges (combined scores ranging 0.400-0.999). The edge list and node list were saved and verified.

Metric	Value	Notes
input_genes	299	DEGs from limma
mapped_genes	272	Mapped to STRING identifiers
unmapped_genes	27	Mostly non-coding RNAs
n_nodes	224	Proteins with ≥ 1 interaction
n_edges	1283	Undirected interaction pairs
required_score_threshold	0.4	Medium confidence (STRING)

1.6 Module Analysis

Module analysis was performed using an MCODE-style dense-module algorithm (Bader & Hogue 2003) with node_score_cutoff=0.5, k_core=2, haircut, and minimum module size 3. Seven modules (size ≥ 3) were detected, covering 172 of 224 nodes (module sizes: 3, 8, 12, 18, 22, 43, 66). The most densely connected module is M01: a fully connected 3-gene module {MT1F, MT1G, MT1H} (3 nodes, 3 edges, density = 1.0000, MCODE score 3.00). All three member genes are down-regulated in UC (MT1F log2FC -1.018; MT1G log2FC -1.981; MT1H log2FC -1.377; all $P < 0.05$). These metallothionein genes were also recovered as a complete 3-node module by Louvain and greedy-modularity community detection, independently corroborating M01. For biological context, the module with the highest MCODE score (density $\times$ size) is M07 (66 genes, density 0.1049, score 6.92), which is chemokine/cytokine/complement-rich (CCL11, CCL18, CCL19, CCL21, CCL24, CXCL11, CXCL13, IL1RL1, IL33R, S100A9, S100A12, TREM1, CFB, CFI, C2, C7, CR1, CD38, GZMK, PPBP, among others).

1.7 Topological Ranking of Network Nodes

All 224 PPI network nodes were ranked using five cytoHubba-style topological algorithms (networkx 3.6.1): MCC (sum over maximal cliques containing the node of $(|C|-1)!$; 547 maximal cliques enumerated), Degree, Closeness (Wasserman-Faust improved), EPC (edge-percolated component, $N=1000$ percolations at edge-keep probability $p=(n-1)/(2m)=0.0869$, seed 42), and MNC (largest connected component of neighbor-induced subgraph). The per-algorithm top 10 genes are summarized below.

Algorithm	Top 10 genes (ranked)
MCC	ICAM1, CXCR4, CXCL9, CXCL10, CXCL8, SELL, CD69, CXCL13, CCL19, CTLA4
Degree	CXCL8 (64), ICAM1 (57), TLR2 (49), CXCL10 (48), CD163 (45), PECAM1/SPP1 (44), CXCR4 (43), TIMP1/CXCL9 (40)

Algorithm	Top 10 genes (ranked)
Closeness	CXCL8 (0.498), ICAM1 (0.476), TLR2 (0.474), SPP1 (0.463), PECAM1 (0.461), TIMP1 (0.452), CXCL10 (0.451), CXCR4 (0.447), CD163/SELL (0.442)
EPC	ICAM1 (75.1), CXCL8 (74.6), CXCL10 (74.0), TLR2 (73.9), CXCR4 (73.4), CD163 (73.0), CXCL13 (73.0), PECAM1 (72.8), CXCL9 (71.8), SELL (71.2)
MNC	CXCL8 (64), ICAM1 (57), TLR2 (49), CXCL10 (48), CD163 (45), SPP1 (43), PECAM1 (42), CXCR4 (42), TIMP1 (40), SELL (40)

The combined ranking table (outputs/files/ranking/topological_scores.tsv) contains 224 gene rows and 14 columns (gene, log2FC, regulation, score+rank per algorithm, avg_rank), with 0 missing scores and sum of degrees 2566 = 2 x 1283 edges. Consistent cross-algorithm hubs include CXCL8, ICAM1, TLR2, CXCL10, CXCR4, PECAM1, CD163, SELL, CXCL9, and CXCL13. The best average ranks were ICAM1 (1.60), CXCL8 (2.00), CXCL10 (4.40), TLR2 (5.40), and CXCR4 (6.00).

Metric	Value	Notes
n_nodes_scored	224	All PPI network nodes
n_algorithms	5	MCC, Degree, Closeness, EPC, MNC
n_maximal_cliques	547	
epc_iterations	1000	Seed 42
epc_keep_probability	0.0869	$p = (n-1)/(2m)$

1.8 Hub Gene Integration and Network Visualization

The five per-algorithm ranks (MCC, Degree, Closeness, EPC, MNC) were integrated for all 224 PPI nodes. The integrated score was defined as the mean over algorithms of the normalized rank $(1 - (\text{rank}-1)/(N-1))$; final ranking was by average rank with competition-style ties. Applying the hub criterion $\text{avg_rank} \leq 10$ yielded 8 final hub genes: ICAM1 (avg_rank 1.60, score 0.9973), CXCL8 (2.00, 0.9955), CXCL10 (4.40, 0.9848), TLR2 (5.40, 0.9803), CXCR4 (6.00, 0.9776), PECAM1 (7.40, 0.9713), CD163 (7.40, 0.9713), and SELL (8.60, 0.9659). All 8 hubs are ranked in the top 20 by all five algorithms ($n_{\text{top20}} = 5/5$) and are up-regulated in UC (log2FC range 1.11-2.47). The hub-induced subnetwork is a complete graph K8: 8 nodes, 28 edges, density 1.0, single connected component. The hub network was visualized with node color and size mapped to integrated score, node border color indicating regulation, and edge width mapped to STRING combined score.

Metric	Value	Notes
n_hub_genes	8	$\text{avg_rank} \leq 10$
n_hub_edges	28	Complete graph K8
hub_subgraph_density	1.0	
n_up_hubs	8	
n_down_hubs	0	
integrated_score_max	0.9973	SELL to ICAM1

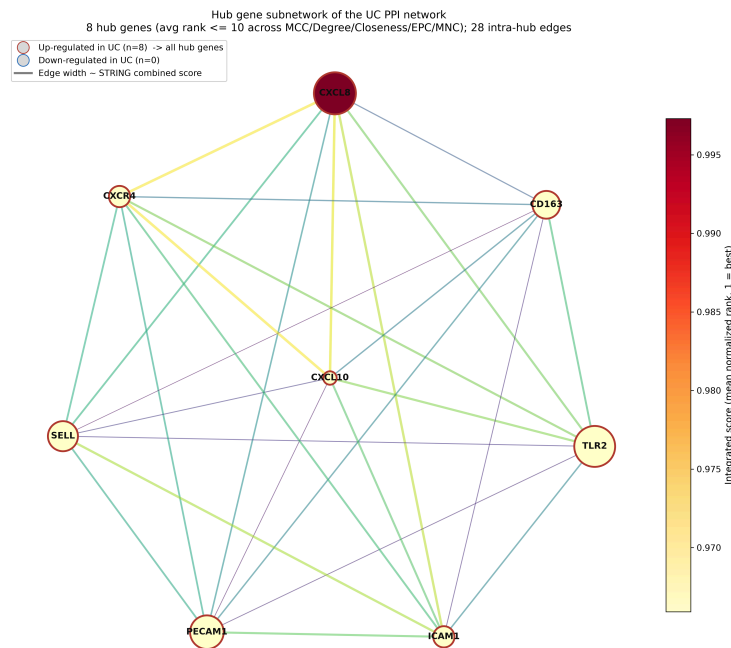


Figure 3 Hub gene network visualization. Node color and size reflect integrated score (YlOrRd colorbar); node border color indicates regulation in UC (red = up, blue = down); edge width reflects STRING combined score.

The 8 hub genes form a complete graph (K8) with 28 edges and density 1.0, indicating strong interconnectivity. All hubs are up-regulated (red borders), and node size/color intensity reflects their integrated importance scores, with ICAM1 and CXCL8 being the most prominent.

1.9 Hub Gene Regulation and Biological Interpretation

All 8 final hub genes (ICAM1, CXCL8, CXCL10, TLR2, CXCR4, PECAM1, CD163, SELL) are up-regulated in active UC relative to healthy controls (log2FC range 1.105-2.473, all $P < 0.05$; 8 up / 0 down), based on the limma log2FC values from the complete DEG list. The annotation table (outputs/files/hub_genes/hub_gene_annotation.tsv) contains 8 rows and 7 columns: gene, log2FC, P_value, regulation_in_UC, functional_category, signaling_pathway, biological_interpretation.

Gene	log2FC	P value	Regulation	Functional category	Signaling pathway
ICAM1	1.157	1.41e-06	Up	Leukocyte adhesion	NF-kB / adhesion cascade
CXCL8	2.473	2.08e-05	Up	Chemokine (neutrophil chemoattractant)	NF-kB / chemokine signaling
CXCL10	1.833	7.69e-04	Up	Chemokine (Th1/Th17, NK recruitment)	CXCR3 chemokine signaling

Gene	log2FC	P value	Regulation	Functional category	Signaling pathway
TLR2	1.105	6.36e-06	Up	Innate microbial sensing	TLR / NF-kB signaling
CXCR4	1.885	1.34e-09	Up	Chemokine receptor (homing/retention)	CXCL12/CXCR4 signaling
PECAM1	1.136	2.72e-07	Up	Leukocyte transmigration	Adhesion / transmigration
CD163	2.110	1.29e-12	Up	Macrophage marker	Tissue remodeling / scavenger receptor
SELL	1.211	1.19e-03	Up	Leukocyte rolling (L-selectin)	Adhesion cascade

Biological interpretation: the hub set forms a coherent pro-inflammatory axis of mucosal immunity. Innate microbial sensing (TLR2) activates NF-kB-driven chemokine/cytokine output (CXCL8/IL-8 neutrophil chemoattractant; CXCL10/IP-10 recruiting CXCR3+ Th1/Th17 and NK cells), feeding a leukocyte adhesion/recruitment cascade from rolling (SELL/L-selectin) to firm adhesion (ICAM-1) to transmigration (PECAM-1). CXCR4 governs CXCL12-directed leukocyte homing/retention, and CD163 marks infiltrating tissue macrophages and chronic tissue remodeling. This coordinated upregulation is consistent with the hallmark histopathology of active UC: neutrophil crypt abscesses, mixed T-cell/macrophage infiltration, and NF-kB/chemokine-driven mucosal inflammation.

Metric	Value	Notes
n_hub_genes	8	
n_up_hubs	8	
n_down_hubs	0	
log2fc_min	1.105	

1.10 Figures

The following figures were generated and verified as part of the analysis pipeline. All figures are included in the final report.

Final Answer: The analysis identified 8 hub genes (ICAM1, CXCL8, CXCL10, TLR2, CXCR4, PECAM1, CD163, SELL) in active ulcerative colitis through differential expression analysis (299 DEGs: 227 up, 72 down), STRING PPI network construction (224 nodes, 1,283 edges), MCODE-style module analysis (most dense module: MT1F/MT1G/MT1H, down-regulated), and integration of five topological algorithms (MCC, Degree, Closeness, EPC, MNC). All 8 hubs are up-regulated in UC and form a coherent pro-inflammatory axis of mucosal immunity: innate microbial sensing (TLR2) drives NF-kB-mediated chemokine output (CXCL8, CXCL10), feeding a leukocyte adhesion/recruitment cascade (SELL rolling, ICAM1 firm adhesion, PECAM1 transmigration), with CXCR4 governing leukocyte homing and CD163 marking infiltrating macrophages. These hubs represent promising candidates for further investigation as biomarkers or therapeutic targets in UC. The complete results, including all figures and tables, are provided in the accompanying PDF report.